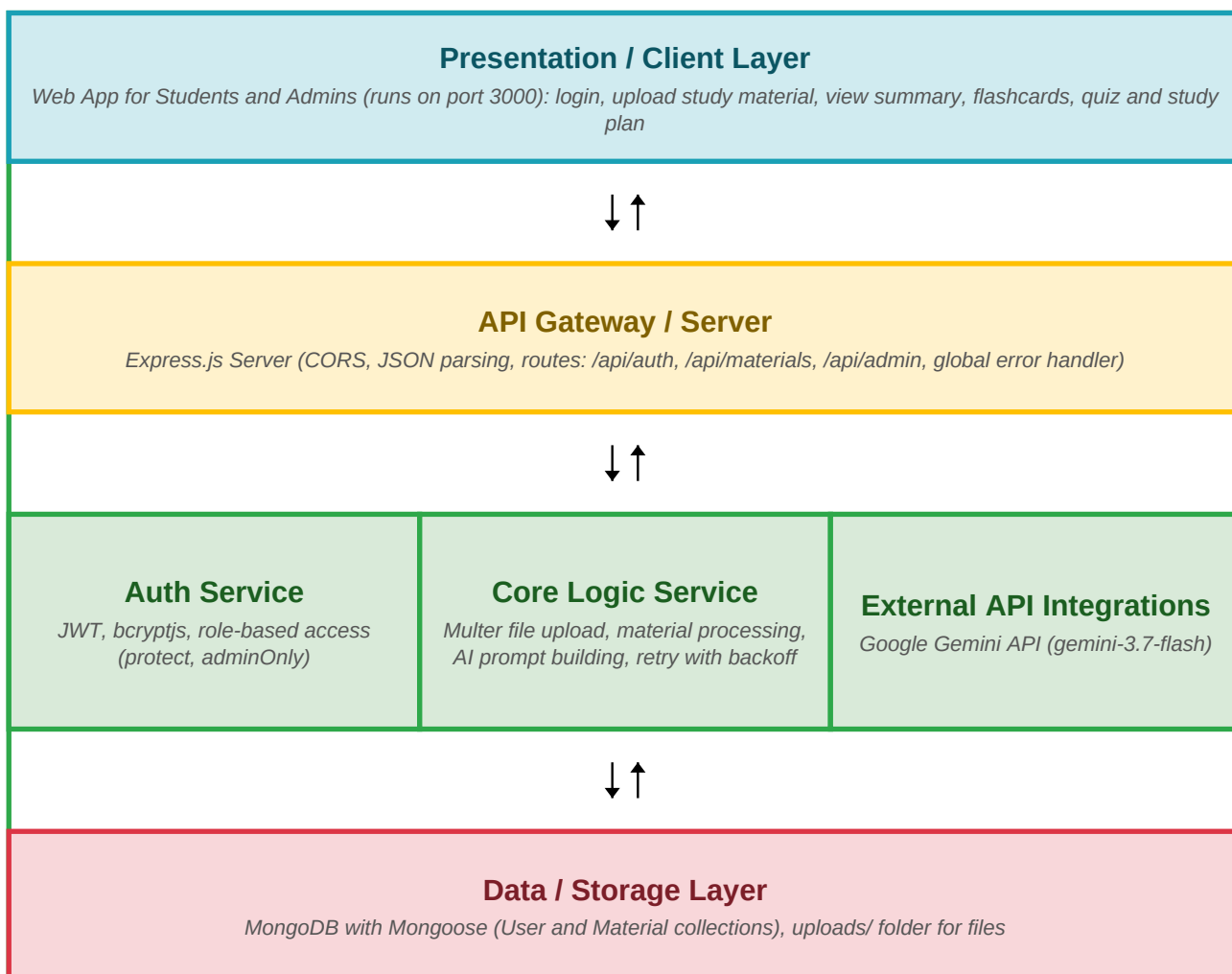


Solution Architecture

Date	30 September 2026
Team ID	xxxxxxx
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Web interface where students and admins log in, upload study files, and view the generated summary, flashcards, quiz and study plan.	Web App (port 3000), REST calls to the API
API Gateway	Entry point of the backend; parses JSON, applies CORS, routes requests to auth, materials and admin endpoints, and handles errors globally.	Node.js, Express.js, cors
Auth Service	Registers and logs in users, hashes passwords, issues and verifies JWT tokens, and restricts admin routes by role.	jsonwebtoken, bcryptjs
Core Logic Service	Accepts .txt/.md/.pdf uploads (max 5 MB), reads the content, builds prompts, calls the AI and saves the results; retries on 503/429 errors.	Multer, Node.js, custom Gemini helper
External API Integration	Generates the summary, flashcards, quiz and study plan from the uploaded material.	Google Gemini API (gemini-3.7-flash)
Database	Stores users (name, email, hashed password, role) and materials (content, summary, flashcards, quiz, study plan) with timestamps.	MongoDB, Mongoose